18. Find the impulse responses of the systems described by these equations.

(a) $4y''(t) = 2x(t) - x'(t)$

(b) $y''(t) + 9y(t) = -6x'(t)$

(c) $-y''(t) + 3y'(t) = 3x(t) + 5x''(t)$

20. A continuous-time function is nonzero over a range of its argument from 0 to 4. It is convolved with a function that is nonzero over a range of its argument from -3 to -1. What is the nonzero range of the convolution of the two?

23. For each graph in figure E.23 select the corresponding signal or signals from the group, $x_1(t)$ to $x_8(t)$.

$x_1(t) = \delta_2(t) * \mathrm{rect}(t/2)$, $x_2(t) = 4\delta_2(t) * \mathrm{rect}(t/2)$, $x_3(t) = (1/4)\delta_{1/2}(t) * \mathrm{rect}(t/2)$

$x_4(t) = \delta_{1/2}(t) * \mathrm{rect}(t/2)$, $x_5(t) = \delta_2(t) * \mathrm{rect}(2t)$, $x_6(t) = 4\delta_2(t) * \mathrm{rect}(2t)$

$x_7(t) = (1/4)\delta_{1/2}(t) * \mathrm{rect}(2t)$, $x_8(t) = \delta_{1/2}(t) * \mathrm{rect}(2t)$,

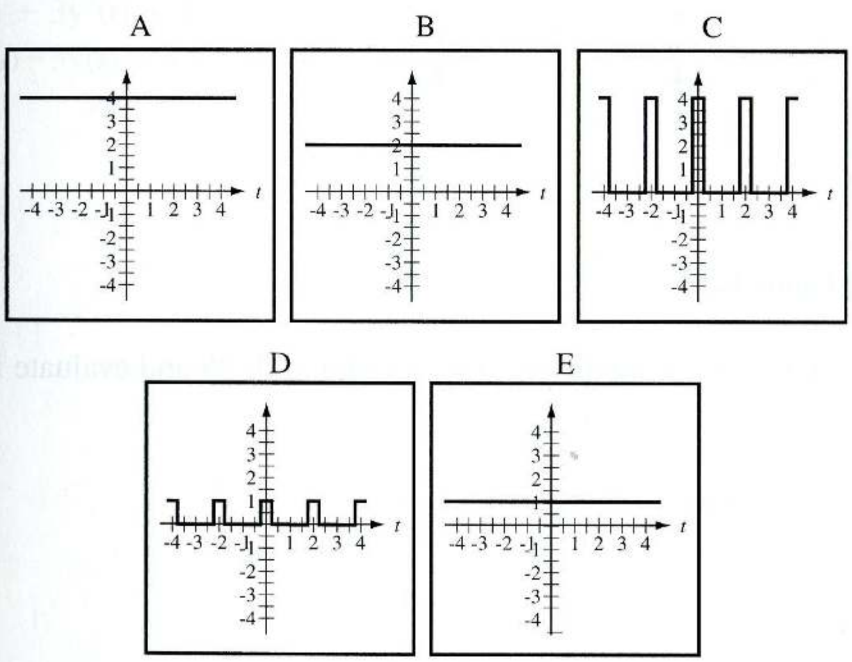
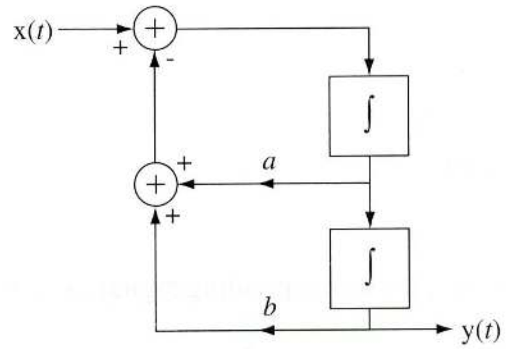


Figure E.23

27. Write the differential equations for the systems in figure E.27, find their impulse responses, and determine whether or not they are stable. For each system, $a = 0.5$ and $b = -0.1$. Then comment on the effect on system stability of redefining the response.

(a)



(b)

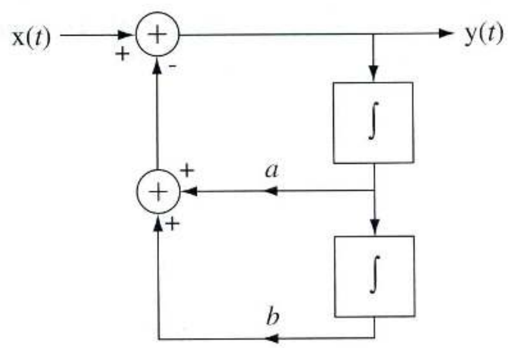


Figure E.27

28. Find the impulse response of the system in figure E.28 and evaluate its BIBO stability.

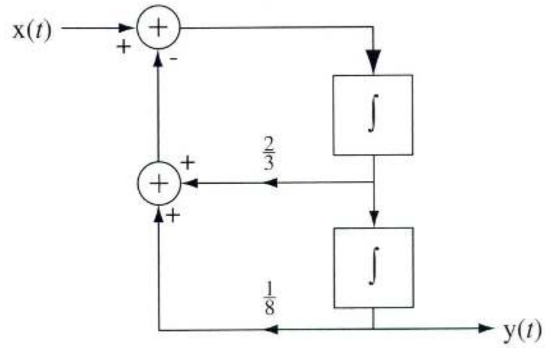


Figure E.28
A two-integrator system.

32. Graph Direct Form II block diagrams of the systems described by these equations. Use integrators, not differentiators, in the block diagrams.

(a) $y''(t) + 3y'(t) + 2y(t) = x(t)$

(b) $4y'(t) - 3y(t) = x''(t) + 2x'(t) - 5x(t)$